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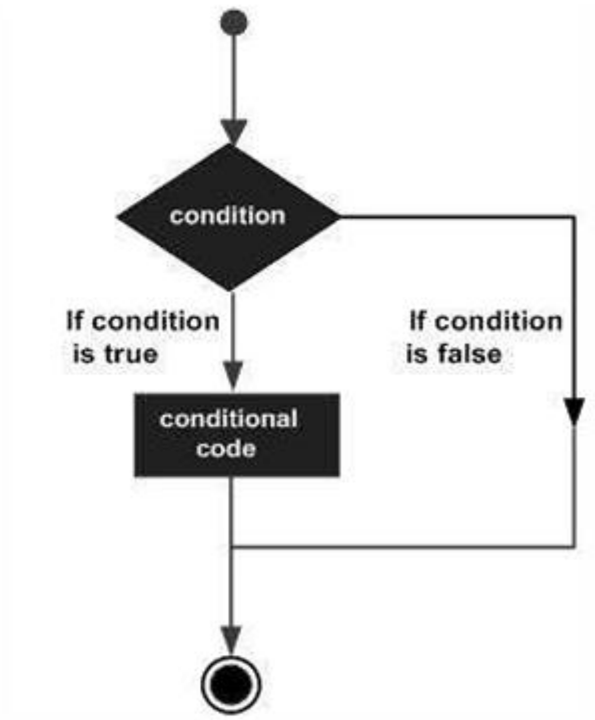
Python If-else statements

Decision making is the most important aspect of almost all the programming languages. As the name implies, decision making allows us to run a particular block of code for a particular decision. Here, the decisions are made on the validity of the particular conditions. Condition checking is the backbone of decision making.

Decision structures evaluate multiple expressions which produce TRUE or FALSE as outcome. You need to determine which action to take and which statements to execute if outcome is TRUE or FALSE otherwise.

Following is the general form of a typical decision making structure found in most of the programming languages –



Python programming language assumes any **non-zero** and **non-null** values as TRUE, and if it is either **zero** or **null,** then it is assumed as FALSE value.

Indentation in Python

For the ease of programming and to achieve simplicity, python doesn't allow the use of parentheses for the block level code. In Python, indentation is used to declare a block. If two statements are at the same indentation level, then they are the part of the same block.

Generally, four spaces are given to indent the statements which are a typical amount of indentation in python.

Indentation is the most used part of the python language since it declares the block of code. All the statements of one block are intended at the same level indentation.

Python has the following decision-making statements:

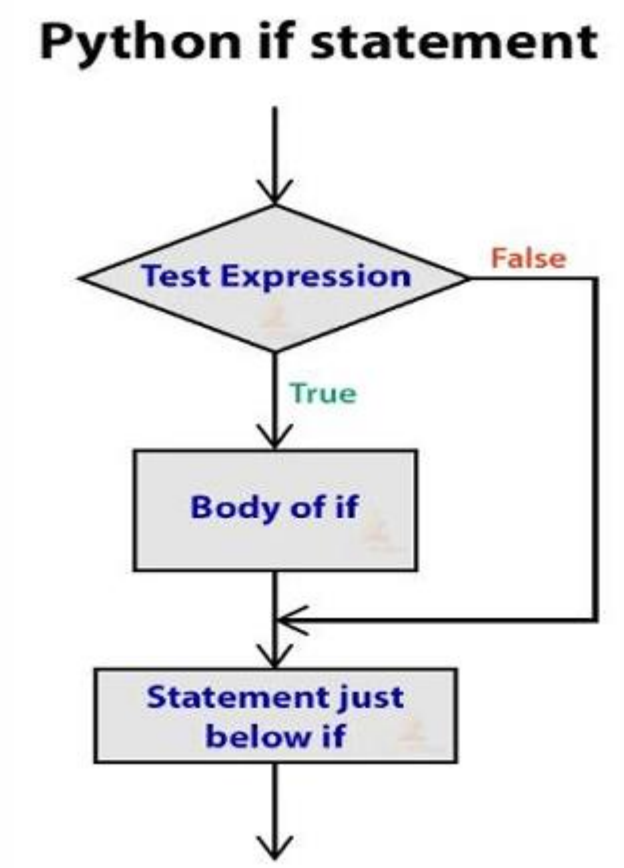
- **if statements**
- **if-else statements**
- **if-elif ladder**
- **Nested statements**

Python if statement

if statement is the most **simple form** of **decision-making** statement. It takes an **expression** and **checks** if the expression evaluates to **True** then the block of code in **if statement** will be **executed.**
If the expression evaluates to **False,** then the **block of code** is **skipped.**

Syntax:

```
if ( expression ):
   Statement 1
   Statement 2
    .
   Statement n
```

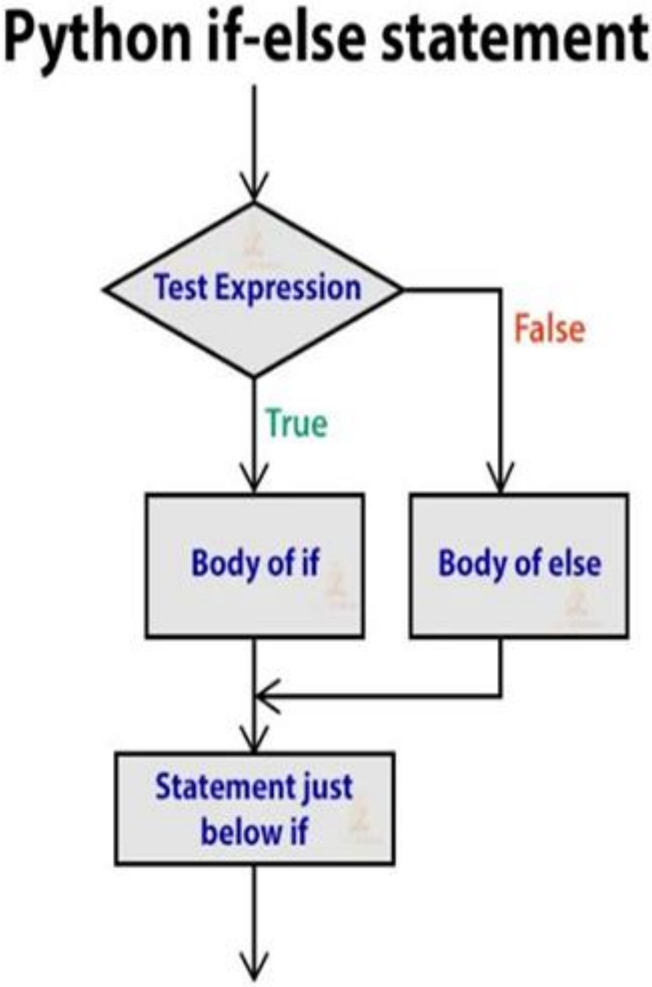


Python if-else statement

From the name itself, we get the clue that the **if-else statement** checks the expression and **executes** the **if block** when the expression is **True** otherwise it will **execute** the **else block of code**. The **else block** should be right after **if block** and it is executed when the expression is **False**.

Syntax:

```
if ( expression ):
Statement 1
Statement 2
.
Statement n
```



Note: Only **one else statement** is followed by **an if statement**. If you use **two else statements** after an **if statement,** then you get the following **error.**

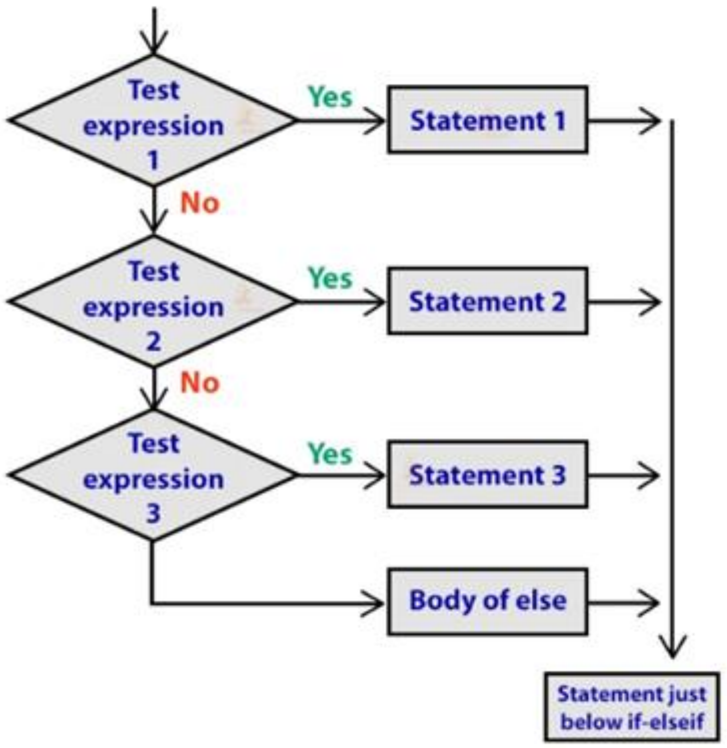
Python if-elif ladder

You might have heard of the **else-if statements** in other languages like **C/C++ or Java.**
In Python, we have an **elif keyword** to chain **multiple conditions** one after another. With elif ladder, we can make complex **decision-making statements**. The elif statement helps you to check **multiple expressions** and it **executes** the code as soon as one of the conditions evaluates to **True.**

Syntax:

```
If ( expression1 ):
    statement
elif (expression2 ) :
     statement
elif(expression3 ):
     statement
  .
  .
  else:
       statement
```

Python if-elif ladder



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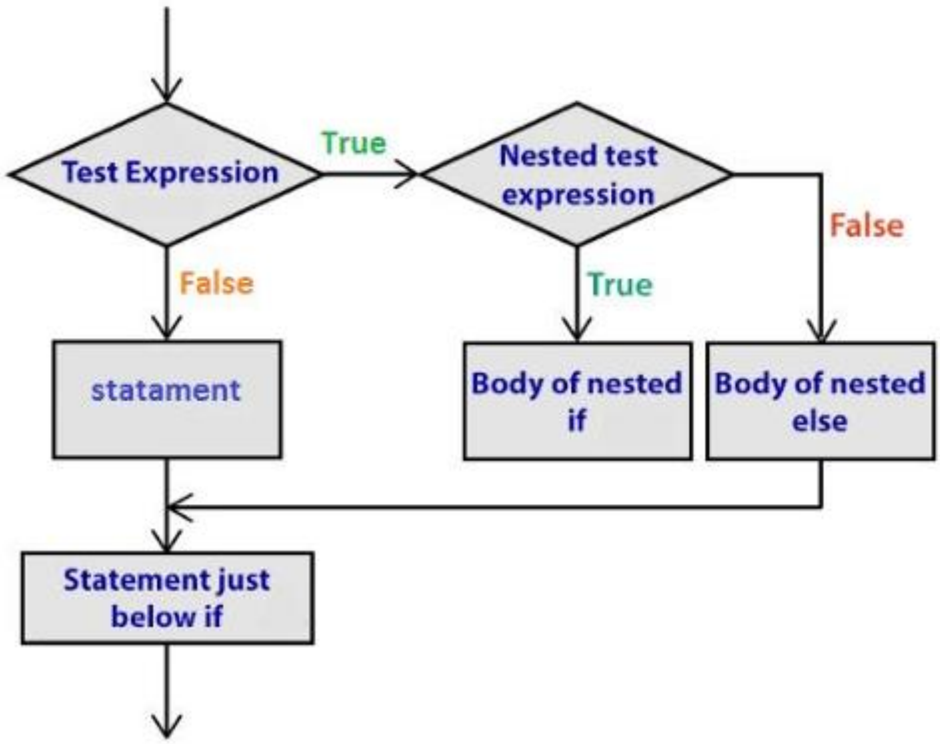
Python Nested if statement

In very simple words, **Nested if statements** is an if statement inside another if statement. Python allows us to **stack** any number of **if statements** inside the **block** of another **if statements**. They are **useful** when we need to make a **series of decisions.**

Syntax:

```
if (expression):
 if(expression):
    Statement of nested if
  else:
    Statement of nested if else
   Statement of outer if
 Statement outside if block
```

Python Nested if statement



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